Negative real zeros in the Ma–Wang alternating-run polynomials

Principia Math In Progress

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Task 1 (MathDB 330839). Ma and Wang's published Conjecture 4.1 [1] asserts that $T_n(x)/x$ has only imaginary zeros and a coordinate-separation property for every $n \geq 2$. Here "imaginary" is read as nonreal, as indicated by the source's use of general conjugate pairs with nonzero real parts. The assertion already fails at $n = 7$; the failure persists for all larger indices.

Theorem 1. *For every integer $n \geq 7$, $T_n(x)/x$ has at least two distinct real zeros in $(-1, 0)$ and at least two distinct real zeros in $(-\infty, -1)$. In particular, the full conjunction in Ma–Wang's Conjecture 4.1 is false.*

Proof. The defining recurrence [1, Eq. (1.1)] is

$$T_{n+1}(x) = (2nx + 1)xT_n(x) + x(1 - x^2)T_n'(x), \qquad T_1(x) = x.$$

Writing $P_n(x) = T_n(x)/x$, we obtain

$$P_{n+1}(x) = \big(1 + x + (2n - 1)x^2\big)P_n(x) + x(1 - x^2)P_n'(x), \qquad P_1 = 1. \tag{1}$$

In particular,

$$P_n(0) = 1, \qquad P_n(-1) = (2n - 3)!! > 0 \quad (n \geq 2). \tag{2}$$

Six applications of (1) give

$$\begin{aligned} P_7(x) = 1 &+ 63x + 1023x^2 + 4641x^3 + 16479x^4 + 26481x^5 \\ &+ 37759x^6 + 26481x^7 + 16479x^8 + 4641x^9 \\ &+ 1023x^{10} + 63x^{11} + x^{12}. \end{aligned}$$

Exact substitution yields

$$P_7(-1/20) = -\frac{316770843280059}{4096000000000000} < 0. \tag{3}$$

Together with (2), the intermediate value theorem already gives two distinct real zeros in $(-1, 0)$.

For persistence, suppose P_n takes a negative value on $(-1, 0)$. By (2), its minimum on $[-1, 0]$ is attained at an interior point a, with $P_n(a) < 0$ and $P_n'(a) = 0$. Equation (1) gives

$$P_{n+1}(a) = \big(1 + a + (2n - 1)a^2\big)P_n(a) < 0,$$

because $-1 < a < 0$ and $n \geq 1$. Its endpoint values are again positive, so it has two distinct zeros in $(-1, 0)$. Induction starting from (3) proves this for every $n \geq 7$.

The source's symmetry assertion [1, Theorem 2.2] gives

$$P_n(x) = x^{2n-2}P_n(1/x).$$

It also follows immediately by induction from the coefficient recurrence in [1, Eq. (1.2)]. Each of the two zeros in $(-1, 0)$ therefore has a distinct reciprocal zero in $(-\infty, -1)$. The four zeros are distinct and nonzero, completing the proof. □

Scope and prior work. This refutation concerns the nonreal-zero assertion in the original conjunction. It is compatible with Hurwitz stability, which permits negative real zeros. Ding and Zhu [2, Conjecture 2.13 and Proposition 2.16] prove Hurwitz stability of the same sequence, while citing Ma–Wang's Conjecture 4.1 for that different statement. Their result is credited and is not contradicted here. Their later journal version, *Advances in Applied Mathematics* 152 (2024), 102591, has not been read in full; its precise discussion of the original conjecture remains a priority-audit gap. This note therefore receives **zero new campaign credit pending that check**. Both complete versions of the original Ma–Wang paper were read.

The accompanying exact verifier derives the displayed coefficients by the recurrence and independently enumerates all 135135 dual Stirling permutations of order seven. Rational sign evaluations and Sturm counts provide additional checks. The persistence argument is analytic and covers infinitely many indices. No independent peer review or formal verification is claimed.

Proposed Tasks 2–5 (not additional solutions).

2. Determine the exact number of negative real zeros of P_n for every $n \geq 7$, including multiplicities.

3. Determine whether the proportion of real zeros of P_n has a limit as $n \to \infty$, and compute it if it exists.

4. Determine the indices for which the real and imaginary coordinates of the nonreal conjugate pairs of consecutive polynomials interlace.

5. Obtain asymptotic formulas for the negative real zeros closest to zero and their reciprocal partners.

References

[1] S.-M. Ma and H.-N. Wang, *Enumeration of a dual set of Stirling permutations by their alternating runs*, Bulletin of the Australian Mathematical Society **94** (2016), 177–186. doi:10.1017/S0004972716000186. Preprint: arXiv:1506.08716v2, 30 June 2015; its corresponding assertion is Conjecture 7.

[2] M.-J. Ding and B.-X. Zhu, *Stability of combinatorial polynomials and its applications*, arXiv:2106.12176v2, 24 June 2021. Related journal version: doi:10.1016/j.aam.2023.102591.